From PDFs to a Skill Tree: A Bayesian Adaptive Mastery System for Arbitrary Course Material

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Abstract. Khan Academy-style mastery learning is effective but hand-authored, and exists for at most a few dozen canonical courses. We demonstrate a working system that generates a Khan-Academy-style mastery experience from arbitrary text-based course material in minutes, and uses a Bayesian adaptive diagnostic to place a learner onto the resulting skill tree before the mastery loop begins. The system contributes (i) a deterministic-with-LLM-fallback pipeline that emits a self-contained offline mastery app from PDFs in a single bring-your-own-key (BYOK) run, (ii) a domain-agnostic Bayesian engine extended for prerequisite-DAG placement via a graph-Laplacian prior, a frontier-stopping criterion, and an asymmetric credential rule, and (iii) a design choice that renders the diagnostic onto the skill tree itself rather than as a separate summary screen. The hosted demo lets a reviewer run the full path in roughly five minutes; three pre-authored courses are available without uploading. Demo: `https://quizvid.vercel.app`. Video: `https://youtu.be/F2JhBMiRQqU`.

Keywords: adaptive testing · mastery learning · Bayesian inference · LLM-generated curricula · skill trees

1 System Overview

The system has two pieces, deployed together as a static gallery plus a serverless authoring path. **Pipeline.** A deterministic parser (PyMuPDF + heading detection, with a multimodal-LLM fallback for scanned PDFs) produces a structured corpus from uploaded course material. A sequence of LLM calls then generates a prerequisite-DAG ontology (auto-selecting single-pass or per-document map-reduce by token volume) and a typed question bank against it; two cheap auxiliary calls enrich the ontology with per-topic spines and a topic-to-source-document chronology that orders branches in the rendered tree. For textbook corpora, a deterministic regex pass detects exercise sections and feeds them to question generation as RAG-style context [2]. The pipeline runs on serverless compute under a BYOK model: the user supplies a Gemini key, the system reports a pre-flight cost estimate, and on confirmation generates a course for roughly five cents (lecture-set) to thirty cents (textbook) in API spend.

Diagnostic engine. Placement on a prerequisite DAG is the system's most novel component. Knowledge tracing literature (BKT [1] and modern variants

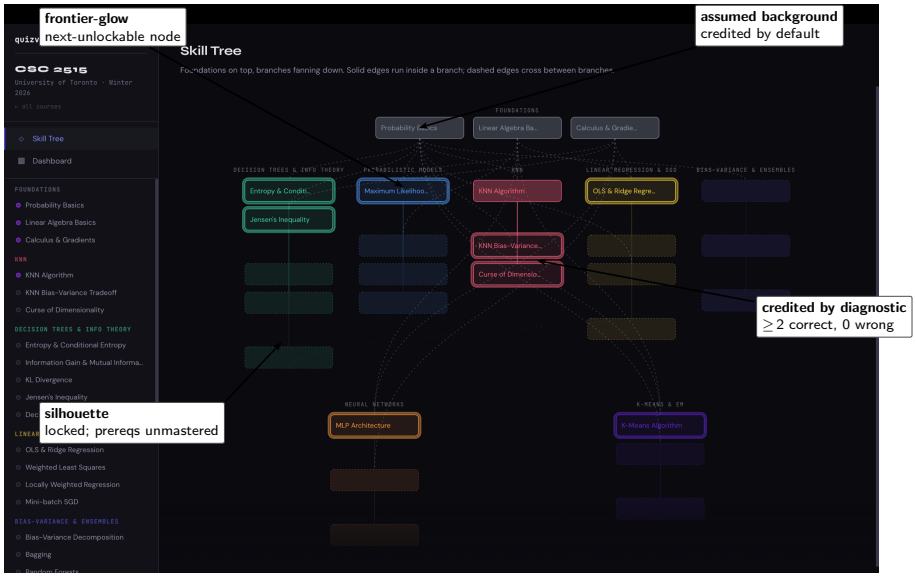

Fig. 1. Skill tree mid-session, csc2515 ML lectures. The diagnostic places the learner directly onto the tree rather than a separate summary screen. Frontier-glow marks next-unlockable nodes; silhouettes mark locked nodes whose prerequisites are not yet mastered; the foundations row is credited as assumed background; the KNN cluster is credited by diagnostic (≥ 2 correct, 0 wrong).

including graph-based extensions [3]) typically targets per-skill mastery across a long sequence of attempts; we instead want *cold-start placement* on a fresh course in tens of questions. We build on a domain-agnostic Bayesian engine originally developed for an MBTI personality quiz, with three changes for the mastery domain. *(1) DAG-aware prior.* $_0 = (I + \lambda L)^{-1}$ where L is the normalized graph Laplacian of the prerequisite DAG, so connected nodes start positively correlated and transitive prerequisites receive correlation weight that decays with graph distance. *(2) Frontier-stopping.* The diagnostic halts when each node's posterior probability of next-unlockable membership is decided, aligning the stop criterion with the user-visible output (the rendered tree). *(3) Asymmetric credential rule.* A node is credited as mastered iff the learner answered ≥ 2 questions on it correctly with zero wrong; the posterior μ is used only for question selection ($\arg\max_i w_i^\top \Sigma w_i$), never as the placement criterion. The first correct answer applies no Bayesian update and triggers a confirming follow-up; any wrong answer disqualifies the node. The pure-guesser false-positive rate is $\approx 5\%$ on a 20-question MCQ-only budget, matched across Python and JavaScript implementations.

2 Demonstration

The demo at `https://quizvid.vercel.app` presents two artifacts above the fold: a gallery of three pre-authored courses (csc2515 ML lectures, cg3207 computer architecture, MAT1856 mathematical finance) and a BYOK upload zone. The three courses were chosen to span the pipeline's working envelope of text-based STEM with explicit prerequisite structure (slide-deck lectures, mixed-content systems courses with diagrams and Verilog, formula-heavy applied math). A reviewer reaches a working mastery loop in two clicks: select a course card, click a node on the rendered tree, answer a question, watch the frontier move (Fig. 1). The placement modal that appears on first visit is dismissible — "Skip placement, browse the tree" seeds an empty mastered-set and exposes the root frontier directly. Self-studiers take placement, which seeds a personalised frontier in 15–25 questions and renders it onto the same tree.

3 Originality and Strengths

We know of no prior open system that takes arbitrary course PDFs and emits a self-contained, offline, diagnostic-aware mastery app in a single BYOK run. Three properties are non-obvious. **Cold-start placement.** Knowledge tracing assumes accumulated learner trajectories; we deliver ∼20-question placement on a fresh course with no prior data, by exploiting the prior covariance from the prerequisite DAG itself. **Diagnostic on the artifact.** The diagnostic is rendered onto the same skill tree the learner will use; their first contact with the tree is already personalised. **Honest BYOK economics.** Cost transparency is enforced before any API call; the system does not store user keys, host user-generated courses, or require sign-in. Limitations are explicit: scoped to text-based STEM with explicit prerequisite structure; cross-course mastery transfer is currently disabled; image-only PDFs need OCR before ingestion.

References

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